Dasari Vijay Ram

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	6.01
SASI EM HIGH SCHOOL	Class XII (CENTRAL BOARD OF SECONDARY EDUCATION)	2023	92.60%
SASI EM HIGH SCHOOL	Class X (BOARD OF SECONDARY EDUCATION ANDHRA PRADESH)	2021	100.00%

POSITIONS OF RESPONSIBILITY

National sports organization (NSO) | IIT HYDERABAD

- Participated in IIT Hyderabad's NSO Badminton program , demonstrating discipline and commitment by regular training.

WORK EXPERIENCE

PLC Automation and HMI Developer | Aztec Process Automation , Hyderabad

- Developed a PLC-based automated bioreactor control system using RSLogix 500 and ladder logic.
- Designed automated sequences for filling, sterilization, fermentation, sampling, and harvesting.
- Implemented timer-based control of motors, valves, and process operations.
- Developed monitoring and alarm logic for pH, temperature, DO, liquid level, and air pressure.
- Designed HMI visualization and process monitoring using FactoryTalk View Studio.
- Implemented emergency-stop, interlocking, and automated mode-transition logic for safe operation.

PROJECTS

3D Simulation of Bertrand's paradox with chord visualization

- Designed a 3D simulation of Bertrand's paradox using two different chord selection methods which allows exploring various geometric probability interpretations.
- Implemented real-time chord rendering using **matplotlib** whose length can be dynamically updated by the user.
- Visualized the paradoxical nature of randomness by comparing outcomes of both methods.

Airline Route Optimizer in C

- Built a graph-based model from real world airline distance data to represent **15,000** nodes using dynamic memory and adjacent matrices.
- Implemented **Dijkstra's algorithm** to compute and display the shortest route between two airports which showcases efficient pathfinding.

PLC Automation & Control Projects

- Practiced PLC I/O mapping, relay logic, timers, counters, internal memory bits, branching, and sequential control, with program verification through RSLogix 500 simulation.
- Built a classroom automation/control system incorporating input sensing, timer/counter operations, and multiple output control conditions.
- Designed a material/object separation system with sensor-based detection, timed sequencing, and actuator control for automated sorting.

- Developed a PLC-based burglar alarm system using interlocking logic and multiple internal control bits to implement security monitoring and alarm activation.
- Implemented a conveyor control system using sensor inputs, motor/actuator outputs, latching logic, timers, and counters for sequential operation.

SKILLS

Programming : C | Python | DSA | JAVA

Libraries : NumPy | Pandas

Other Softwares : Solid Edge | Rockwell RSLogix 500 | FactoryTalk View Studio (SCADA)

RELEVANT COURSES

Computer Science	Introduction to Programming | Fundamentals of Machine Learning | Artificial Intelligence | Data Structures and Applications | Algorithms | Optimization -1 | Operating Systems | Theory of Computation | Computer Networks | Introduction to Computing
Mathematics	Calculus I | Calculus II | Discrete Mathematics for Computer Science | Elementary Linear Algebra | Differential Equations | Probability and Random Variables | Complex Variables | Introduction to Statistics
Other Courses	Molecular and Cellular Biology | Biological Engineering | Numerical Methods | Digital Fabrication Lab | Fundamentals of Organizational Structures | Maths for Physics | Modern Physics | Physics of Solids | Fluid Mechanics | Nucleic Acids and RNA Biomedicine | Transform Techniques | Engineering Drawing

EXTRACURRICULAR

- Love playing football and badminton in my leisure time.
- Avid supporter of FC Barcelona with keen interest in club's performance and tactics.
- Enjoy playing FIFA online using strategy and quick decision-making , driven by a strong passion for football.